AUSELIA

ADS7057 test board

A 3.3 V board that lives on the ZedBoard PMOD, one step at a time.

The topology, the datasheet facts that matter, the routing of the board, and what each part is and where it sits.

Version 0.3 / October 15, 2026 / Nicolás Villegas

How to read this manual

Read the text first. Then look at the figure. The figure shows the same thing with numbers.

01 **One panel, one step**

A page holds four steps in a 2 × 2 grid. Each panel has a title, a text and a figure.

02 **Example tag**

A figure with this tag is a worked example. Its numbers come from the netlist, the board file or the simulation.

03 **Source**

The KiCad project in hw/, the simulations in sim/, the vendor datasheets and one real JLCPCB quote.

04 **Estimate**

A number marked ESTIMATE is not measured. Read it as a size, not as a promise.

05 **Names**

R16 is a reference designator. It is the same name in the schematic, the BOM and the silkscreen file.

06 **Order**

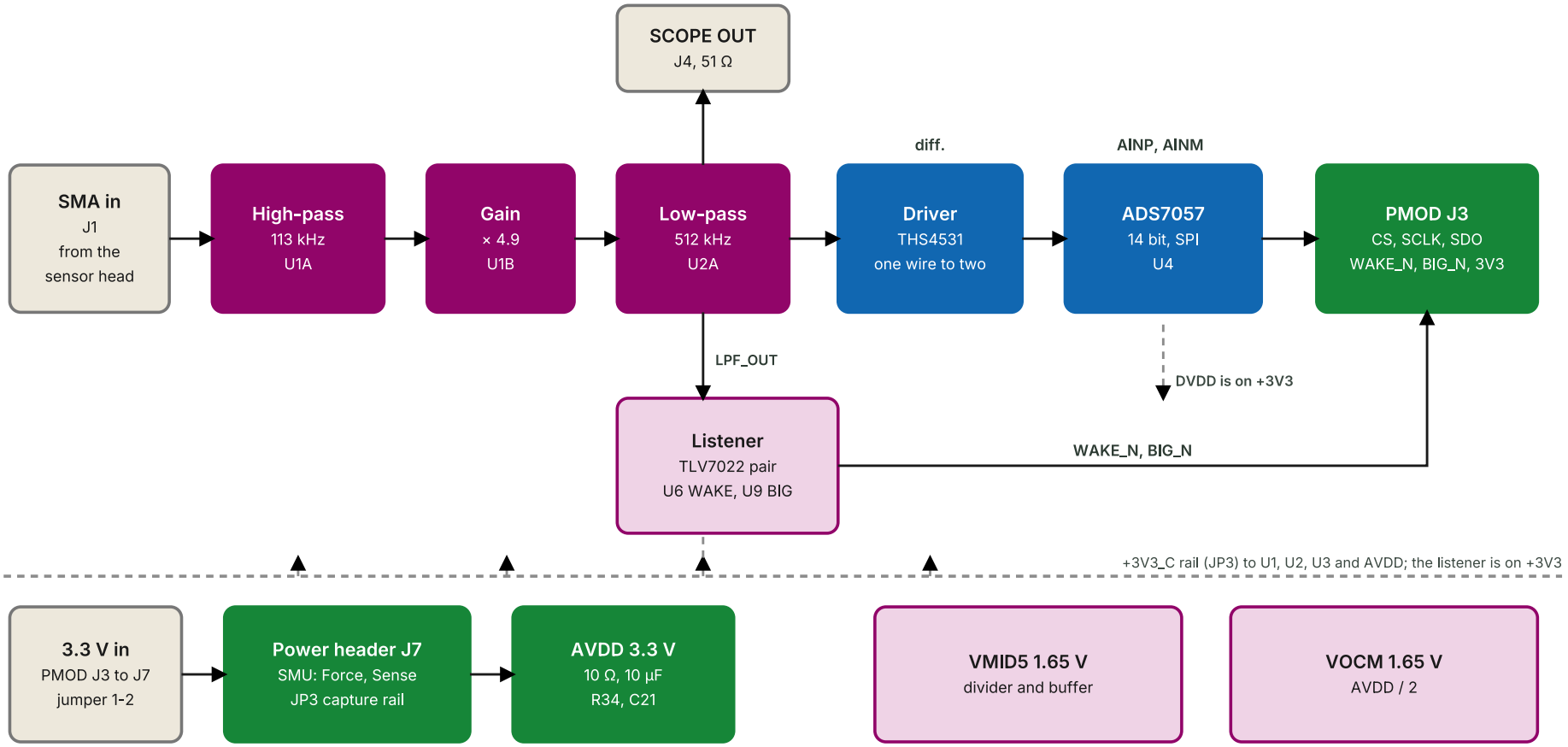
Part 0 gives the basics. Steps 1 to 54 follow the power, the signal, the ADC, SPI, the wake-up, the board and the bring-up.

Colors

signal chain · driver and ADC · power · control, data of the example · the item of this step

The board in one view

the signal goes left to right, the power runs along the bottom



THE BASICS

part 0

What the signal is, why 1.65 V, what an ADC measures and why the filter comes first.

POWER

steps 1 to 6

3.3 V from the PMOD only, the power header J7, the AVDD filter and the 1.65 V mid-rails.

SIGNAL AND ADC

steps 7 to 24

High-pass, gain, low-pass, then the driver and the ADC with its input filter.

SPI AND WAKE-UP

steps 25 to 35

The SPI wires, the frame and its power-up rule, and the listener that raises WAKE_N and BIG_N.

THE BOARD

steps 36 to 48

Layers, floor plan, every part and where it sits, routing, files for the factory and the cost.

BRING-UP

steps 49 to 54

Power-on checks, the first signal, the expected response and what is still open.

PART 0

The basics

0.1 TO 0.6

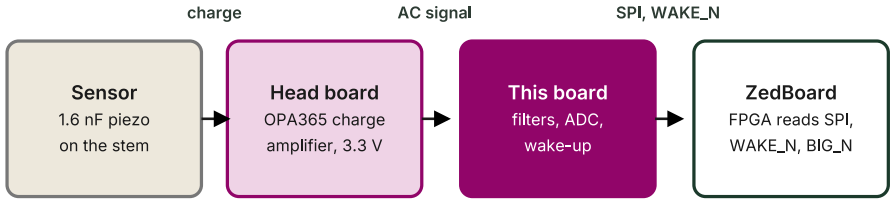
What the board is for, what the signal looks like, why everything rests at 1.65 V, what an ADC measures, why the filter comes first and how to read a PCB. No board details yet.

0.1 THE BASICS
What the board is for

A piezoelectric sensor on a plant stem hears the clicks of **acoustic emission** (AE). The perfboard charge amplifier turns the charge of the sensor into a voltage.

This board takes that voltage, filters it and measures it with a 14-bit ADC. It hands the numbers to the ZedBoard over SPI. A small listener watches the same signal and raises WAKE_N and BIG_N on the PMOD when a click is strong. The whole board runs from the ZedBoard 3.3 V.

CHAIN



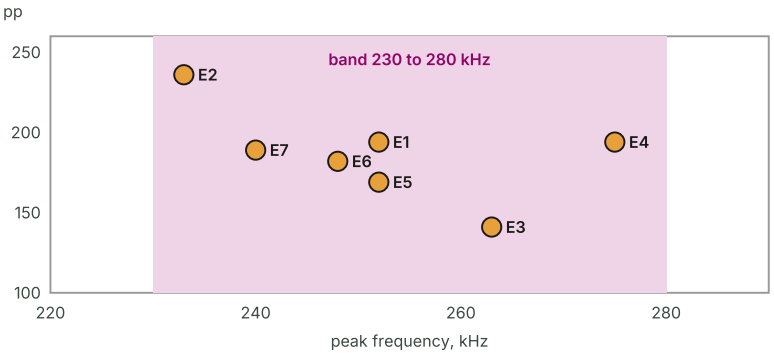
the gray box is the part that this manual describes

0.2 THE BASICS
What the signal looks like

The perfboard output rests near 4.8 V. An AE event rides on it as a burst between 233 and 275 kHz that lasts 26 to 83 µs. The seven recorded events are 141 to 236 mV peak to peak.

Between events, five of the eight baseline records show 3.7 to 4.6 mV rms of noise in the band. That noise sets the smallest event that this board can see.

7 RECORDED EVENTS

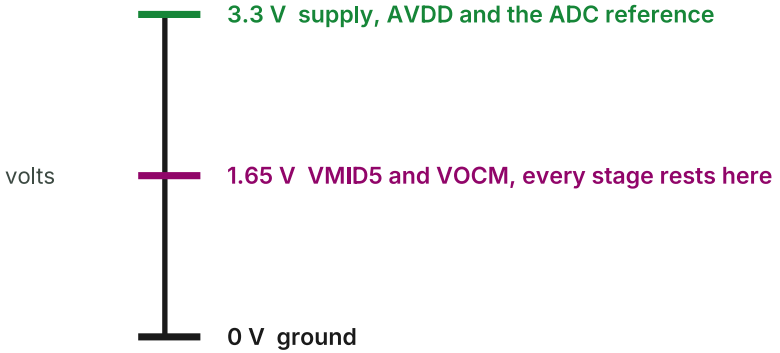


0.3 THE BASICS
Why everything rests at 1.65 V

This board has no negative supply. A signal that swings below 0 V would be cut off. So each stage rests at the middle of its own supply, and the signal swings around that level.

The whole board runs from one 3.3 V rail. The op-amps, the driver and the ADC inputs all rest at half of it, **1.65 V**. The sensor head and the listener rest there too.

VOLTAGE LADDER

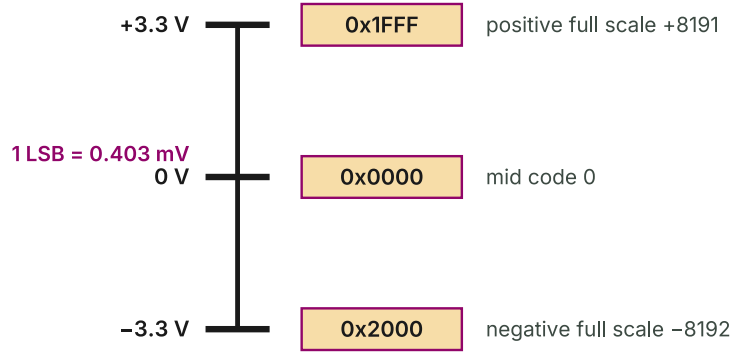


0.4 THE BASICS
What an ADC measures

The ADS7057 measures the **difference** between two pins, AINP and AINM. It reports the difference as a 14-bit signed number in two's complement. One step, called 1 LSB, is 0.403 mV.

The full-scale range is plus and minus the supply AVDD, so ±3.3 V. The code 0x0000 means no difference.

CODE LADDER

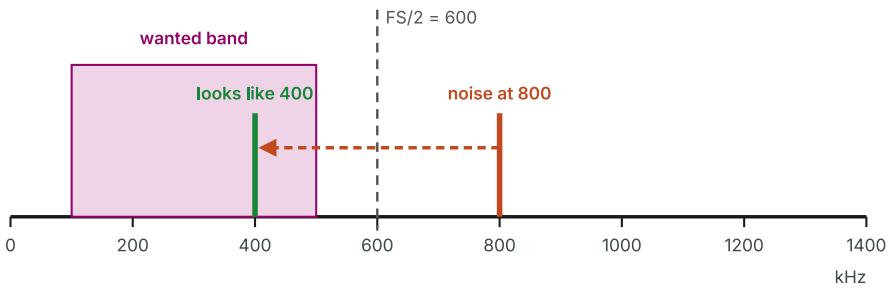


0.5 THE BASICS
Why filter before sampling

A sampled signal cannot tell a frequency f from FS – f. At 1.2 MSPS a tone at 800 kHz looks exactly like a tone at 400 kHz, inside the band of interest.

So a low-pass filter must remove what lies above half the sample rate before the ADC sees it. The sample rate here is about 1.2 MSPS with a 25 MHz SPI clock (ESTIMATE).

ALIASING

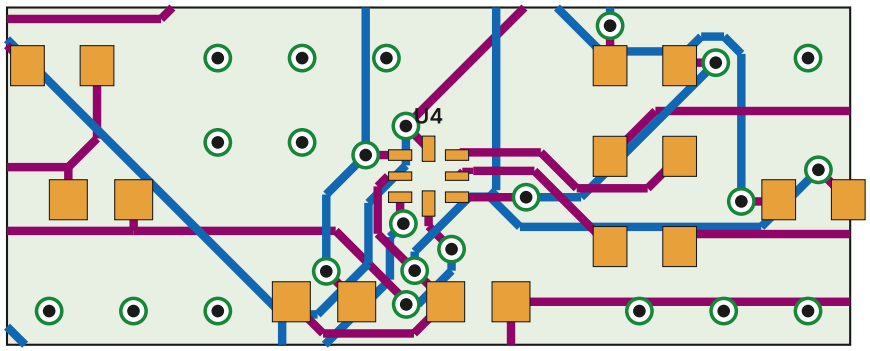


0.6 THE BASICS
How to read a PCB

The board has two copper layers. A **pad** holds a part. A **track** joins pads. A **via** is a plated hole that joins the two layers. A **pour** fills free space with copper, here ground.

The **silkscreen** prints the names. The figure is a 20 × 8 mm cut of this board around the ADC.

REAL BOARD, 6 ×



amber: pads purple: top tracks blue: bottom tracks green ring: vias pale green: ground pour

NOTES

NOTES

SECTION 1

Power and references

STEPS 1 TO 6

How 3.3 V comes in from the ZedBoard PMOD alone, how the power header J7 and the link JP3 serve it, how AVDD is filtered and how the 1.65 V mid-rails are made. Everything else on the board rests on these rails.

BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

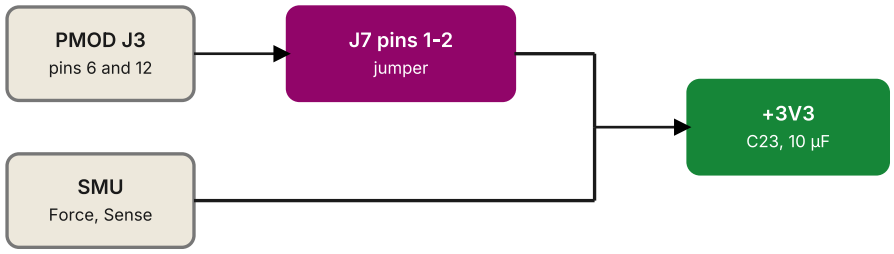
POWER AND REFERENCES

1 How 3.3 V comes in

The board takes all its power from the ZedBoard PMOD. Pins 6 and 12 of J3 carry 3.3 V, and pins 5 and 11 carry ground. There is no USB, no regulator and no diode. The 5-pin header J7 is the power entry: pin 1 is the PMOD 3.3 V, pins 2 and 3 are the board rail +3V3, pins 4 and 5 are ground. A jumper from pin 1 to pin 2 powers the board from the ZedBoard.

For a bench supply or a source-measure unit (SMU), remove the jumper. Connect Force to pins 2 and 5, and Sense to pins 3 and 4. Never use the jumper and the SMU together.

INPUT PATH



one source at a time: jumper or SMU (pins 2 and 3)

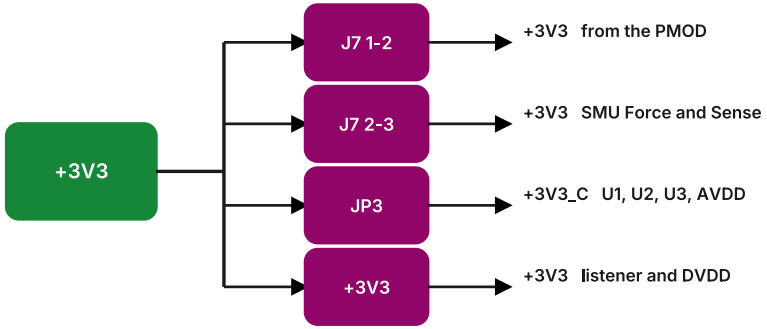
POWER AND REFERENCES

2 The power header J7 and the link JP3

J7 has five pins for a four-wire SMU: 1 Onboard +, 2 Force +, 3 Sense +, 4 Sense -, 5 Force -. Pins 2 and 3 meet at the board rail. Pins 4 and 5 are ground. The SMU reads the board current.

JP3 is a solder jumper, closed, between +3V3 and the capture rail +3V3_C (U1, U2, U3 and AVDD). The listener and DVDD sit on +3V3. Open JP3 to measure the capture chain alone or to feed it from a switch later.

RAILS AND LINKS



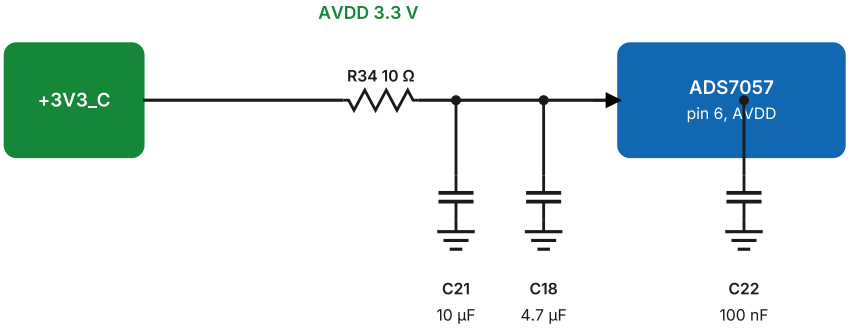
POWER AND REFERENCES

3 Where the ADC reference comes from

The ADS7057 uses its supply pin AVDD as its reference. Full scale is ±AVDD, so noise on AVDD is noise in every code.

There is no second regulator. AVDD comes from the capture rail through a 10 Ω resistor (R34) and 10 µF (C21). That RC has a corner at 1592 Hz and removes 44 dB at 250 kHz (ESTIMATE). The 4.7 µF C18 and the 100 nF C22 sit at the ADC pin.

AVDD PATH



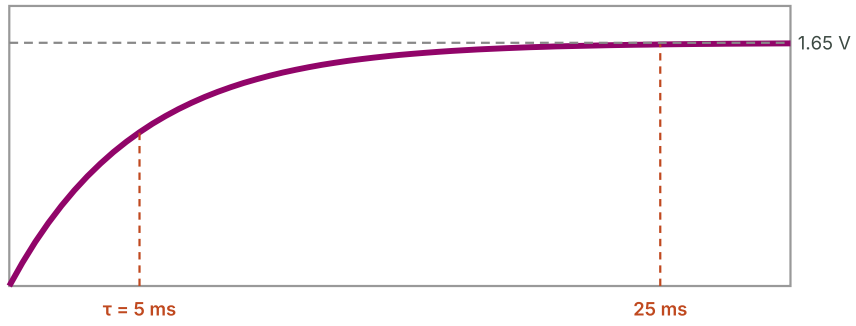
POWER AND REFERENCES

4 How the 1.65 V mid-rail VMID5 is made

Two 10 kΩ resistors (R7, R8) divide the capture rail +3V3_C in half, 1.65 V. The 1 µF capacitor C7 smooths it. The op-amp U2B then copies this voltage and drives the rail called **VMID5**.

The divider and C7 form an RC of 5 ms, so VMID5 needs about 25 ms to settle after power-up. R9 (10 Ω) and C8 (10 µF) keep the buffer calm.

SETTLING AFTER POWER-UP



time after power-up

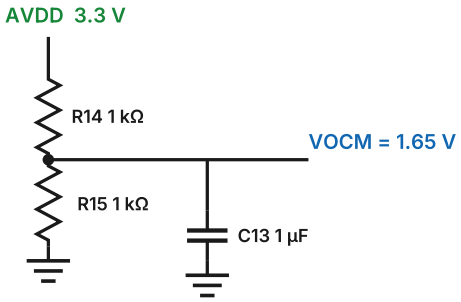
POWER AND REFERENCES

5 How the 1.65 V mid-rail VOCM is made

Two 1 kΩ resistors (R14, R15) divide **AVDD** in half, and C13 smooths the result. This voltage, VOCM, tells the THS4531 where to centre its two outputs. It is a second 1.65 V rail, separate from VMID5.

The datasheet asks for the ADC inputs to rest at AVDD/2, within ±0.1 V. The divider follows AVDD by ratio, so any change of AVDD moves VOCM with it.

VOCM



allowed window

1.55 to 1.75 V

AVDD/2 ± 0.1 V

POWER AND REFERENCES

6 How much current, and where to probe

The board draws about 26 mA from the 3.3 V of the PMOD (ESTIMATE from datasheet typical values). The four op-amp channels and the sensor head are the largest parts. The listener is only about 25 µA.

Every stage has a test point. The list below names the rails. J7 and JP3 (step 2) let you read the total current with an SMU, or the capture chain alone.

ESTIMATE, mA

- OPA2365, 4 channels 18
- Sensor head 4.7
- VOCM divider 1.6
- ADS7057 AVDD 1.1
- THS4531 0.25
- VMID5 divider 0.17
- Listener, 2 × TLV7022 0.022

+3V3	TP11	**AVDD**	TP12	**VMID5**	TP13
VOCM	TP14	**LST**	TP15	**GND**	TP19, TP20

NOTES

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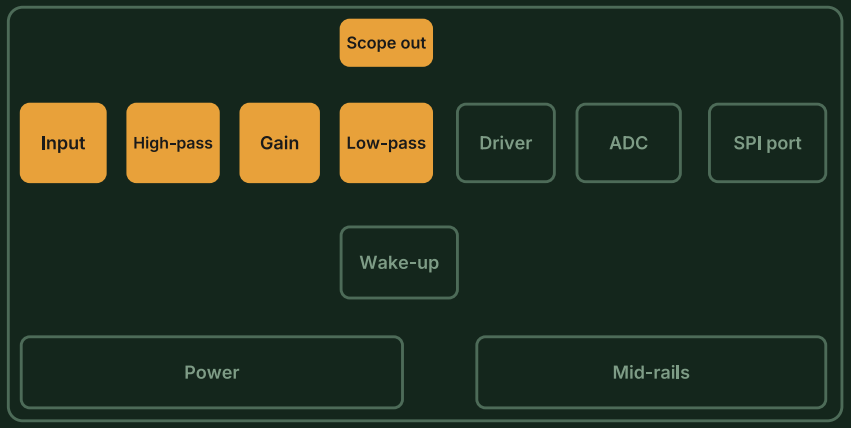
SECTION 2

The signal path

STEPS 7 TO 16

From the SMA connector to the output of the low-pass filter: input coupling, high-pass, gain, low-pass, the total response, headroom, the SCOPE OUT tap and the op-amps.



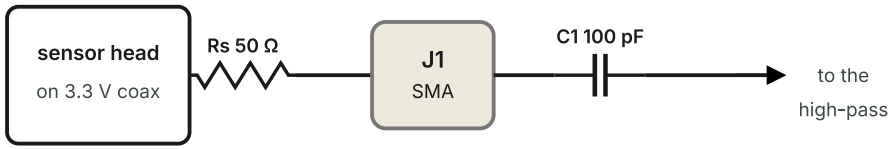
BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

7 THE SIGNAL PATH
How the signal gets in

The sensor head reaches J1 through a coax cable. The head has its own 49.9 Ω output resistor. The same cable carries the 3.3 V that powers the head, so the centre pin of J1 sits near 3 V.

So the first part is a 100 nF capacitor, C33. It passes the 250 kHz burst and blocks the DC level. A 1 MΩ resistor R27 to VMID5 sets the DC level of the next node, and C1 continues to the high-pass.

INPUT



at 250 kHz, C1 is 6.4 kΩ. At DC it is open.

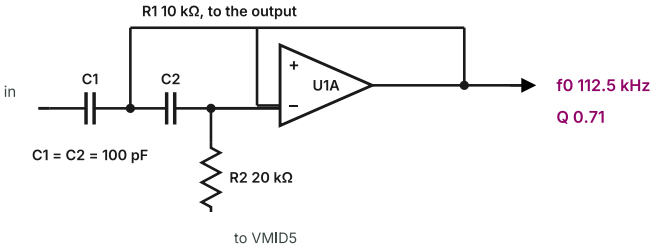
8 THE SIGNAL PATH
How the high-pass works

The high-pass is a **Sallen-Key** stage: two capacitors in series, two resistors and one op-amp used as a buffer. Low frequencies cannot pass the capacitors.

$$f_0 = \frac{1}{2\pi C\sqrt{R_1 R_2}}, \qquad Q = \frac{1}{2}\sqrt{\frac{R_2}{R_1}}$$

With C = 100 pF, R1 = 10 kΩ and R2 = 20 kΩ, the corner is 112.5 kHz. Q is 0.71. That is a maximally flat response.

SALLEN-KEY HIGH-PASS

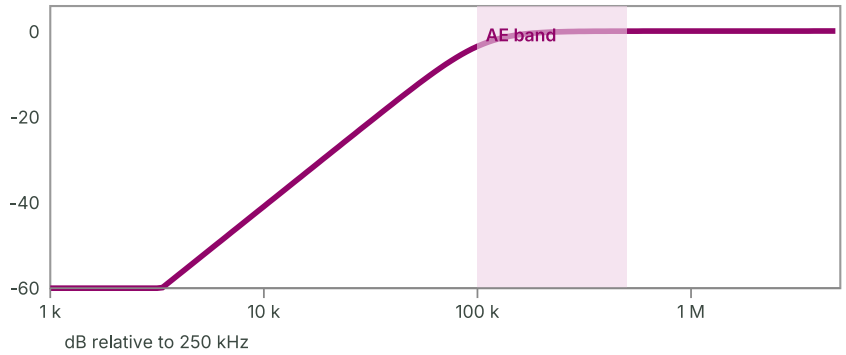


9 THE SIGNAL PATH
Why the corner is at 113 kHz

Most noise on the perfboard sits below 100 kHz, from 11.5 to 26.8 mV rms. The AE band starts at 100 kHz and holds only about 4 mV rms.

So the high-pass sits just under the band. At 10 kHz it already removes 41 dB of this stage alone. A corner any higher would also cut the events.

SIMULATION, HIGH-PASS STAGE ONLY



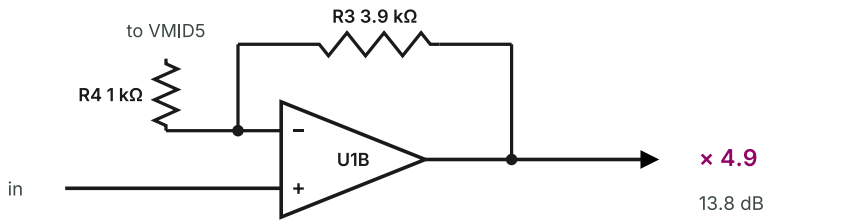
10 THE SIGNAL PATH
How the gain is set

The second half of U1 is a non-inverting amplifier. Its gain comes from two resistors: R3 from the output back to the minus input, and R4 from that input to VMID5.

$$G = 1 + \frac{R_3}{R_4} = 1 + \frac{3.9\,\text{k}\Omega}{1\,\text{k}\Omega} = 4.9$$

Because R4 goes to VMID5 and not to ground, the gain is 1 for DC. The stage output rests at 1.65 V, and the gain applies only to the burst.

GAIN 4.9



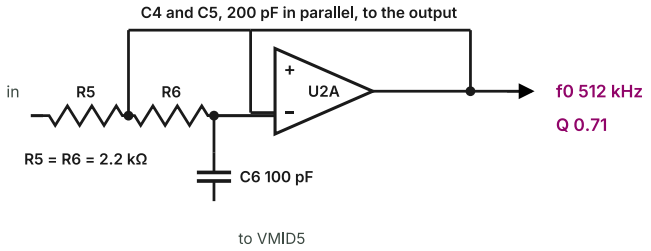
11 THE SIGNAL PATH
How the low-pass works

The low-pass is also a Sallen-Key stage, with the parts swapped: two resistors in series and two capacitors. The capacitors give high frequencies a path to VMID5.

$$f_0 = \frac{1}{2\pi R\sqrt{C_1 C_2}}, \qquad Q = \frac{1}{2}\sqrt{\frac{C_1}{C_2}}$$

With R = 2.2 kΩ, C1 = 200 pF (C4 and C5 in parallel) and C2 = 100 pF (C6), the corner is 512 kHz. Q is 0.71.

SALLEN-KEY LOW-PASS

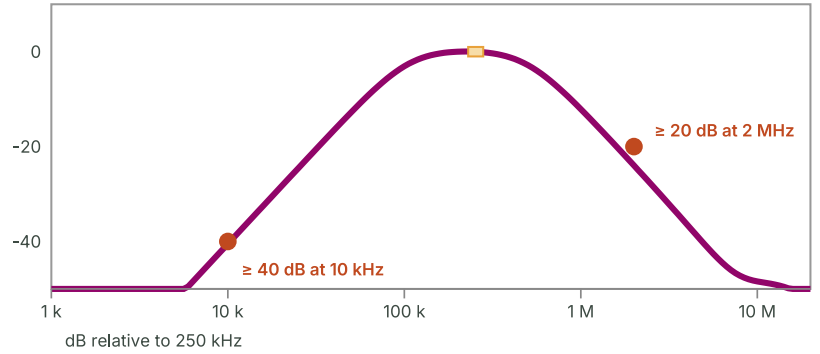


12 THE SIGNAL PATH
The total response

Put the three stages together and add the 50 Ω source. The simulation, with TI's own OPA365 model, gives 12.3 dB of gain at 250 kHz. That is ×4.1, a little below the ideal ×4.9.

The loss at 10 kHz is 40.5 dB against a target of 40 dB. With part tolerances 11 of 150 runs fall under 40 dB, the worst at 39.7 dB.

SIMULATION, SMA TO ADC

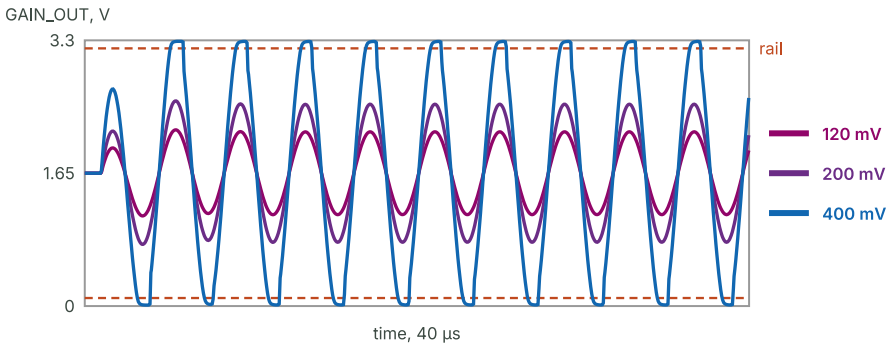


13 THE SIGNAL PATH
What clips first

The 3.3 V stages are the narrowest part of the path. They can swing about 1.55 V either side of 1.65 V. A burst clips there when it is larger than 0.35 V at the sensor head output.

A typical recorded event is 120 mV, so the margin is 2.9×. At the ADC that event fills 16 % of the full scale.

SIMULATION, 3 BURSTS

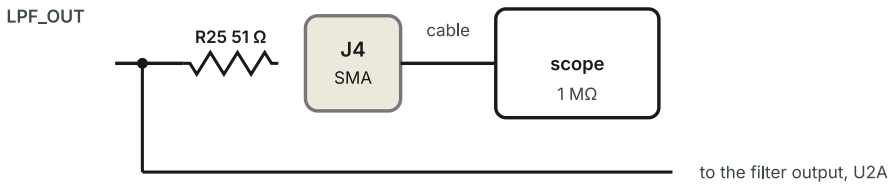


14 THE SIGNAL PATH
What SCOPE OUT is for

J4 gives the scope the same signal that the ADC sees, taken at the output of the low-pass. A 51 Ω resistor (R25) sits between the op-amp and the connector.

The resistor keeps the cable capacitance away from the op-amp, so the filter cannot ring. Use the 1 MΩ input of the scope. A 50 Ω input would form a divider with R25 and halve the signal.

CABLE LOAD



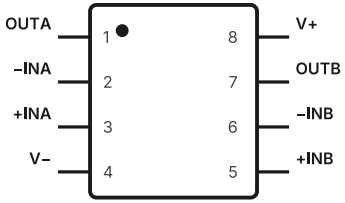
1 MΩ input: 1.000 of LPF_OUT. 50 Ω input: 0.50

15 THE SIGNAL PATH
What the OPA2365 does here

Two dual op-amps, U1 and U2, do all the active filtering. The OPA2365 is rail-to-rail on input and output and runs from 2.2 to 5.5 V. Its gain-bandwidth is 50 MHz and its noise is 4.5 nV/√Hz.

Each package holds two channels. The table shows which stage uses which channel.

SOIC-8, TOP VIEW



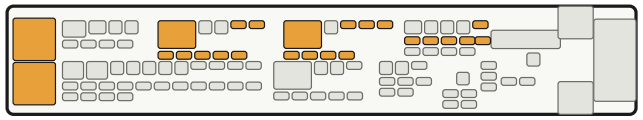
U1A pins 1 2 3	high-pass buffer
U1B pins 5 6 7	gain × 4.9
U2A pins 1 2 3	low-pass buffer
U2B pins 5 6 7	VMID5 buffer

16 THE SIGNAL PATH
Where these parts are

The signal parts sit in the top half of the board, from J1 on the left to U2 on the right. U1 holds the high-pass and the gain. U2 holds the low-pass and the VMID5 buffer.

J4 and its resistor R25 stand in the top right corner, away from the ADC.

TOP VIEW, ACTIVE PARTS IN AMBER



Input and high-pass
C1 C2 R1 R2

Gain
R3 R4

Low-pass
R5 R6 C4 C5 C6

Op-amps
U1 U2 C3 C10

VMID5
R7 R8 C7 R9 C8 C9

Connectors
J1 J4 R25

NOTES

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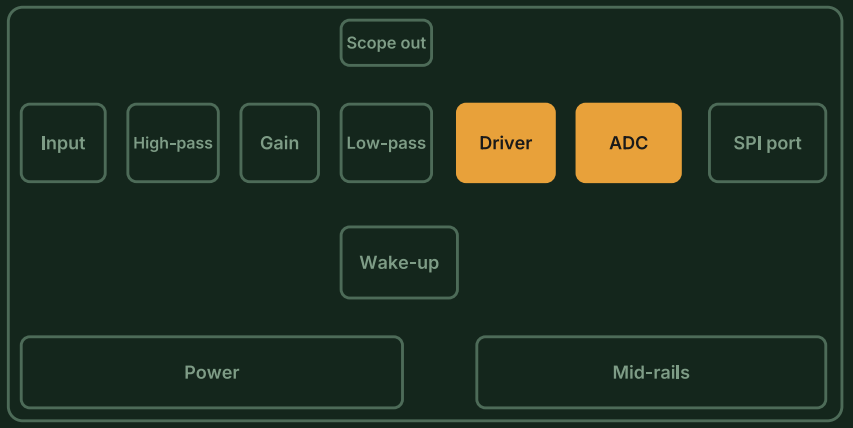
SECTION 3

Driver and ADC

STEPS 17 TO 24

How one wire becomes two, how the ADC input is protected from its own sampling kick, what the ADS7057 is and what its numbers mean for this board.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

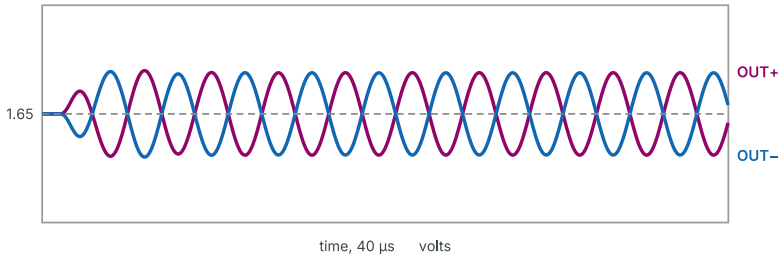
17 DRIVER AND ADC
How one wire becomes two

The ADS7057 needs two inputs that move in opposite directions. The **THS4531** (U3) makes them from one wire. The capacitor C11 and R10 bring the signal in. The corner of this coupling is 159 Hz, far below the band.

$$V_{OUT+} - V_{OUT-} = \frac{R_{12}}{R_{10}} V_{IN} = 1 \cdot V_{IN}$$

Each output moves half of the signal around VOCM, in opposite directions.

SIMULATION, 120 mV BURST



18 DRIVER AND ADC
What to know about the THS4531

Three facts from the datasheet shape the board. The **PD** pin must be driven, so U3 ties it to 3.3 V. The **VOCM** pin accepts 0.95 to 2.3 V on a 3.3 V supply, so 1.65 V is fine.

The third fact is a limit. The designers of the part expect about 2 pF of load between its outputs. A larger load needs a series resistor.

DATASHEET AGAINST THIS BOARD

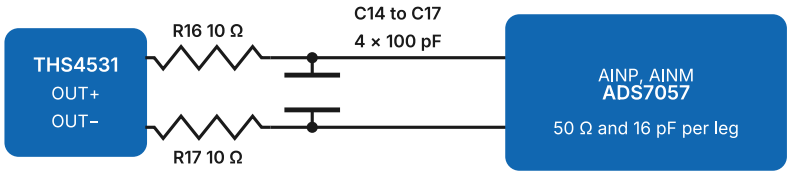
item	datasheet	this board	
Supply	2.5 to 5.5 V	**3.3 V**	
Gain-bandwidth	27 MHz	**gain 1**	
Noise	10 nV/√Hz	**0.2 LSB**	
VOCM range at 3.3 V	0.95 to 2.3 V	**1.65 V**	
PD pin	must be driven	**tied to 3.3 V**	
Load between outputs	2 pF nominal	**400 pF + 20 Ω**	OPEN

19 DRIVER AND ADC
What the 10 Ω and 400 pF are for

When the ADC samples, its 16 pF capacitor connects to the input and pulls a burst of charge. A step of 0.56 V moves 9 pC. A big capacitor next to the pins supplies this charge, so the driver sees only a small glitch.

Four 100 pF capacitors make 400 pF, which is 25 times the sampling capacitor. The datasheet asks for at least 20 times. The two 10 Ω resistors isolate the driver. The pole is at 20 MHz, far above the band.

KICKBACK FILTER



the node dips by 22 mV (53 LSB) and recovers in 54 ns at worst

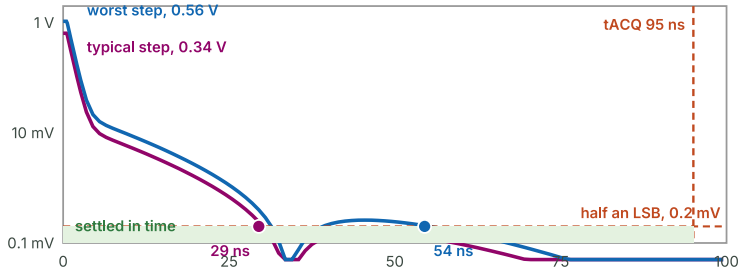
the 25× rule: 16 pF × 20 = 320 pF minimum

20 DRIVER AND ADC
What the ADC input looks like

The sampling switch of the ADC is 50 Ω in series with 16 pF on each leg. The simulation closes the switch on a signal that was left at the previous sample. The input settles to half an LSB in 29 ns for a typical step and 54 ns for a large one.

How to read the graph: at time 0 the switch closes and the input is wrong by the step (340 or 560 mV). The curve is the remaining error as it shrinks. When it falls under the red line, the sample is right to half an LSB. The ADC allows 95 ns.

SIMULATION, SETTLING



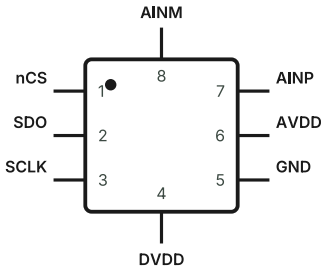
DRIVER AND ADC

21 What the ADS7057 is

A 14-bit successive-approximation ADC in a 1.5 × 1.5 mm package with eight pads. It samples on the falling edge of CS and sends its result over SPI.

The table lists the numbers that matter for this board. All come from datasheet SBAS821.

X2QFN-8, TOP VIEW



Resolution	14 bit, no missing codes
Rate	up to 2.5 MSPS
Supply	AVDD 2.35 to 3.6 V, DVDD 1.65 to 3.6 V
Input	differential, ±AVDD, common mode AVDD/2
SNR	79.6 dB at 2 kHz
INL, DNL	±0.9 LSB, ±0.3 LSB typical
SPI	mode 1, SCLK up to 60 MHz
Conversion	18 SCLK, then at least 95 ns acquisition

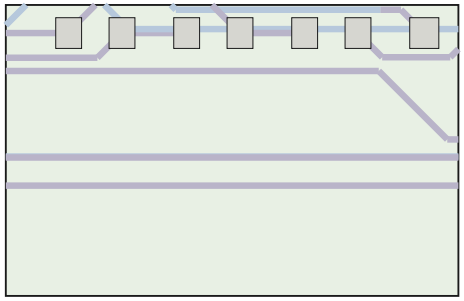
DRIVER AND ADC

22 How the ADC is decoupled

AVDD gets a 4.7 µF (C18) and a 100 nF (C22). DVDD gets a 1 µF (C19) and a 100 nF (C20). The datasheet asks for 3.3 µF and 1 µF and says to keep them close, with no vias between the pin and the capacitor.

On this board C22 is 2.4 mm from the ADC and C18 is 2.7 mm. C20 is 4.5 mm and C19 is 4.7 mm. The 100 nF parts sit farther than the large ones, which is a point to improve.

REAL BOARD, 7 ×



AVDD parts green, DVDD parts blue

DRIVER AND ADC

23 How a voltage becomes a code

The 120 mV event of the simulation reaches 0.517 V between AINP and AINM. The code is the voltage divided by 1 LSB, rounded down.

$$\text{code} = \left\lfloor \frac{V_{\text{AINP}} - V_{\text{AINM}}}{1\ \text{LSB}} \right\rfloor$$

So the peak is 1284, which is 0x0504. The negative peak -0.507 V gives -1258, which is 0x3B16 in two's complement.

FROM THE SIMULATION

D13 is the sign bit

	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	
positive peak	0	0	0	1	0	1	0	0	0	0	0	1	0	0	0x0504
negative peak	1	1	1	0	1	1	0	0	0	1	0	1	1	0	0x3B16

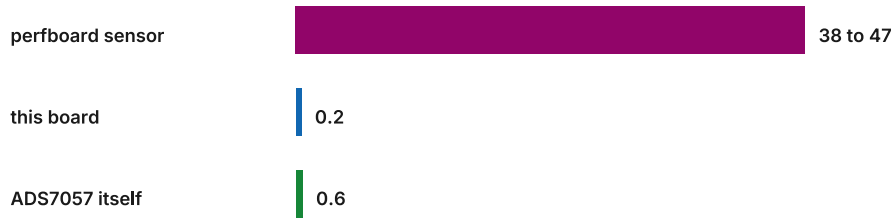
DRIVER AND ADC

24 How much noise the ADC sees

Three noise sources add. The sensor noise of the perfboard, through this board, is 38 to 47 LSB rms. The board adds 0.2 LSB. The ADC itself adds about 0.6 LSB.

So the sensor decides the noise floor. The board and the ADC are small by comparison, and this is what a test board should show.

LSB RMS, ESTIMATE FOR THE ADC

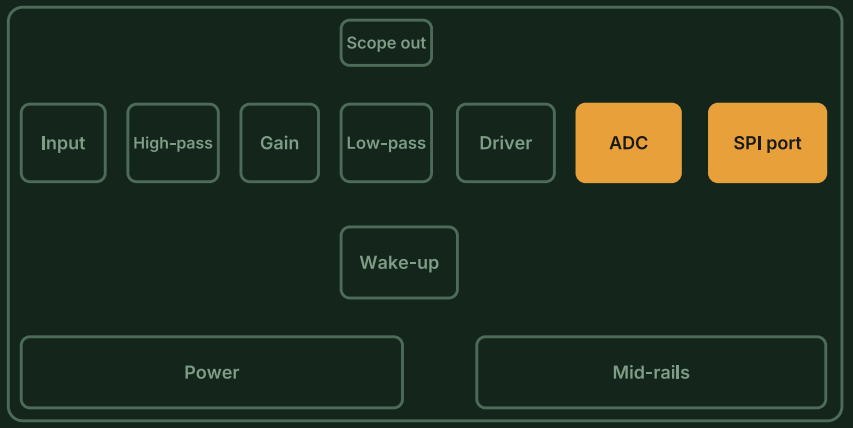


SECTION 4

The SPI link

STEPS 25 TO 30

The four wires from the ADC to the ZedBoard, what one frame looks like, why the data comes one frame late, the calibration at power-up and how fast the link can go.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

25 THE SPI LINK
Which pin is which on the PMOD

The board plugs into one PMOD port of the ZedBoard, JA to JD. The board uses nine of the twelve pins. **CS**, **SCLK** and **SDO** carry SPI. **WAKE_N** and **BIG_N** are the two listener lines. 3V3 on pins 6 and 12 powers the whole board, and GND is the return.

There is no MOSI wire, because this ADC only sends data. The numbering follows the Digilent SPI PMOD, so pin 1 is CS.

J3, SEEN FROM THE TOP

left column	right column
1 CS	7 BIG_N
2 not used	8 not used
3 SDO	9 not used
4 SCLK	10 WAKE_N
5 GND	11 GND
6 3V3	12 3V3

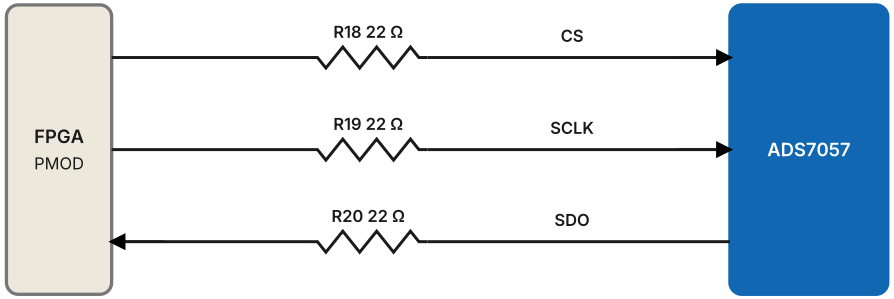
The number is the PMOD pin. Row 1 holds pin 1 and pin 7.

26 THE SPI LINK
Why 22 Ω in each line

R18, R19 and R20 sit in the CS, SCLK and SDO lines at the ADC. Together with the small capacitance of the trace and the pin they slow the edge of the FPGA output a little and damp ringing.

The cost in speed is tiny. 22 Ω with about 10 pF (ESTIMATE) gives 0.2 ns, and one bit at 60 MHz lasts 16.7 ns.

SERIES RESISTOR

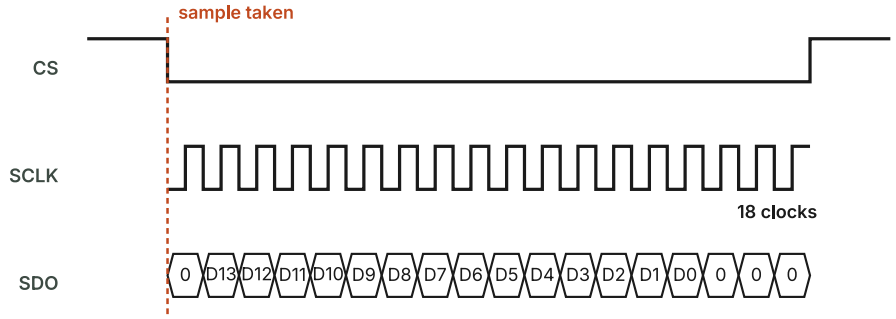


27 THE SPI LINK
What one frame looks like

A **frame** starts when CS falls. The ADC samples at that moment. The host then sends 18 clock pulses. On SDO the ADC answers with a leading 0, then 14 data bits, most significant bit first, then three zeros.

After the 18th clock the ADC converts nothing more. CS goes high and the ADC acquires the next sample for at least 95 ns.

ONE FRAME

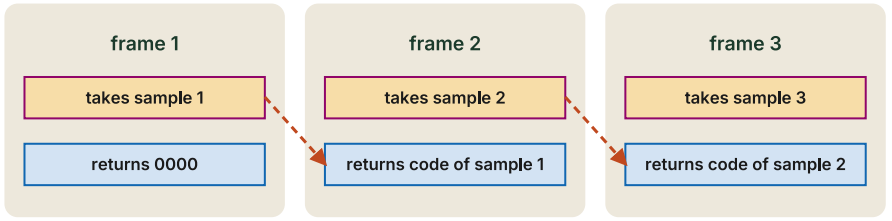


28 THE SPI LINK
Why the data is one frame late

The ADC uses the clocks of a frame to send the result of the **previous** conversion. The conversion of the current sample runs inside the same frame, and its result waits for the next one.

So the first frame after power-up returns all zeros, and every later frame returns the sample that was taken one frame before.

THREE FRAMES

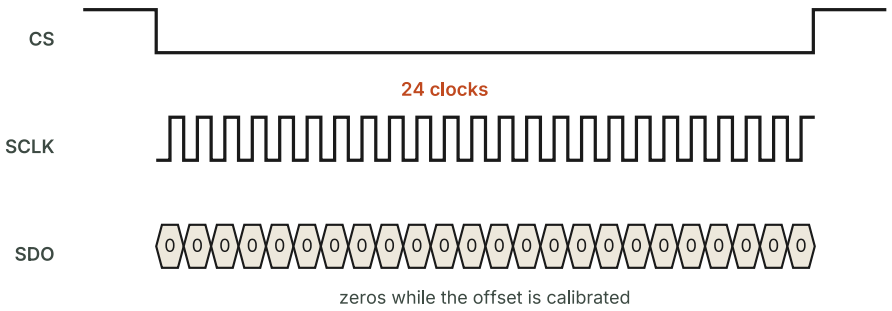


29 THE SPI LINK
What happens at power-up

The datasheet recommends an offset calibration at power-up. The host does it by sending **24** clocks in the first frame, not 18. The ADC then disconnects its input, corrects its own offset and sends zeros.

At a 20 MHz clock this first frame takes 1200 ns plus 95 ns of acquisition, so 1295 ns. Repeat it when AVDD or the temperature changes a lot.

FIRST FRAME



30 THE SPI LINK
How fast the link can go

One sample costs 18 clock periods plus 95 ns. So the sample rate is set by SCLK. The datasheet allows up to 60 MHz, which gives the full 2.5 MSPS.

There are no level shifters on this board, so the FPGA drives the ADC pins directly at 3.3 V. Start at 10 to 20 MHz and raise the clock when the first test works.

SAMPLE RATE AGAINST SCLK

SCLK	one sample	sample rate
10 MHz	1895 ns	**0.53 MSPS**
20 MHz	995 ns	**1.01 MSPS**
25 MHz	815 ns	**1.23 MSPS**
40 MHz	545 ns	**1.83 MSPS**
60 MHz	395 ns	**2.50 MSPS**

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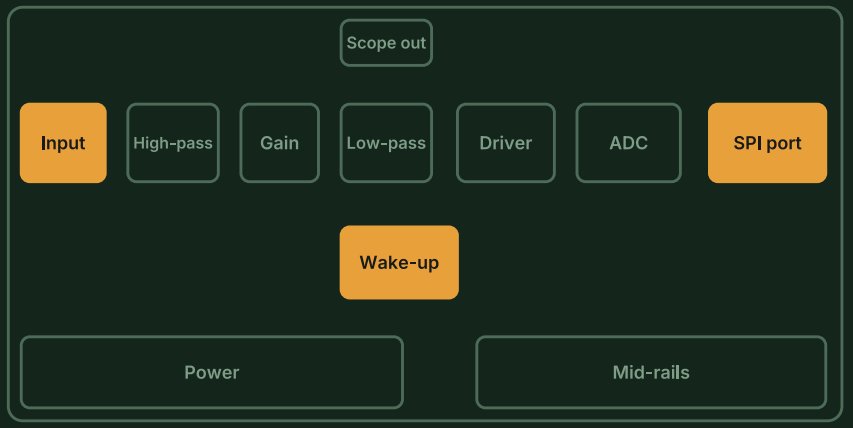
SECTION 5

The listener

STEPS 31 TO 35

How a small circuit next to the capture chain raises WAKE_N and BIG_N when a burst arrives, how the thresholds are set by one resistor chain and how to test them on the bench.



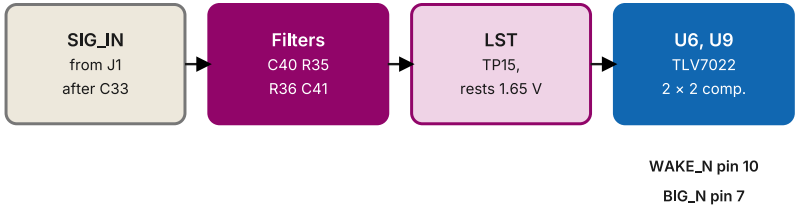
BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

31 THE LISTENER
What the listener does

The listener is a second, very small circuit that watches SIG_IN, the same signal the chain sees. A 100 pF and 10 kΩ high-pass (159 kHz) and a 10 kΩ and 22 pF low-pass (723 kHz) keep only the AE band. The two filters pass 0.68 of the signal at 250 kHz. Two comparators judge the result.

It is an **observer**. It never switches the capture chain. The chain stays on, so the FPGA has the full waveform and the listener decisions side by side. Both outputs are open-drain lines on the PMOD: WAKE_N for a medium burst and BIG_N for a large one.

LISTENER, SIG_IN TO PMOD



comparator inputs are LST against the ladder of step 32

32 THE LISTENER
How the thresholds are set

One resistor chain from 3.3 V to ground makes all four thresholds. Its middle node REFL rests at 1.65 V. Each comparator pair compares LST with a node above and a node below REFL, so either polarity trips it.

$$V_{\text{half}} = I \cdot R, \qquad I = \frac{3.3\,\text{V}}{2.096\,\text{M}\Omega} = 1.57\,\mu\text{A}$$

Divide by the filter gain 0.68 to refer LST to SIG_IN. Charge uses Cf = 47 pF.

THRESHOLDS, FROM THE RESISTORS

Output	Resistors	At LST	At SIG_IN	Charge, Cf 47 pF
WAKE_N (U6)	R39, R40 15 kΩ	24 mV	35 mV	1.6 pC
BIG_N (U9)	R22 + R39, 48 kΩ	76 mV	111 mV	5.2 pC

Chain: R21 1 MΩ, R22 33 kΩ, R39 15 kΩ, REFL, R40 15 kΩ, R41 33 kΩ, R23 1 MΩ.

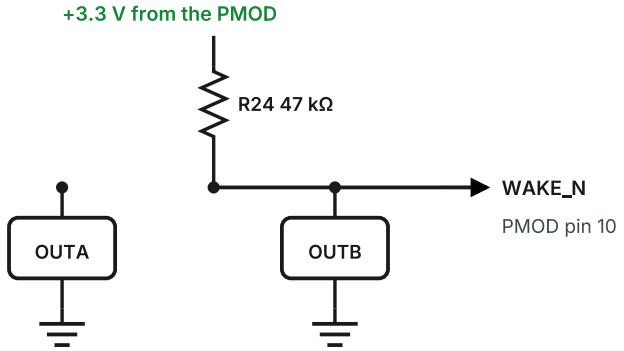
With the OPA314 head (Cf 220 pF) volts per pC fall by 4.7. The ladder then needs changing, because the thresholds approach the comparator offset (plan, section 13).

33 THE LISTENER
Why WAKE_N and BIG_N are open-drain

The TLV7022 output stage is one transistor. It can pull the output to ground. It cannot drive it high. A 47 kΩ resistor (R24 for WAKE_N, R42 for BIG_N) pulls the line up to 3.3 V.

The two comparators of one package share one wire, so either one pulls the line low. The FPGA sees a clean 3.3 V level and never more.

WIRED-OR



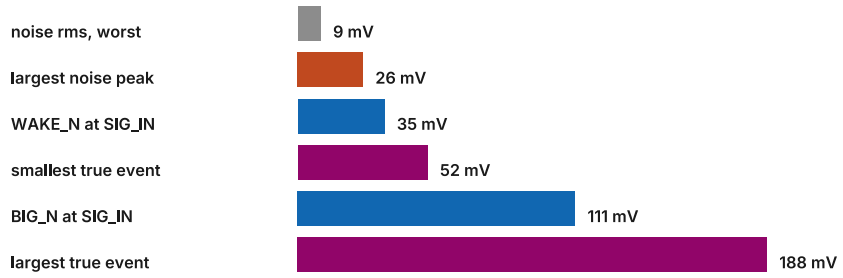
OUTA	OUTB	WAKE_N
off	off	high
on	off	low
off	on	low
on	on	low

34 THE LISTENER
How fast and how accurate

The TLV7022 answers in about 260 ns and draws 5 µA per channel. Its input offset is up to ±8 mV and it adds about 7 mV of hysteresis. Against a half window of 24 mV at LST that is a third, so treat each threshold as ±8 mV at LST, about ±12 mV at SIG_IN.

Real data decide the setting. On the 25 recorded captures the noise after the 100 kHz high-pass is 4.0 mV rms (worst 9.2 mV). The largest of the 18 noise captures peaks at 26 mV, under the 35 mV of WAKE_N at SIG_IN, a margin of only 1.3×. The 7 true events peak at 52 to 188 mV, all above it (ESTIMATE, scaled to Cf 47 pF).

THRESHOLD AGAINST DATA, MV AT THE HEAD



THE LISTENER

35 How to test the listener

Feed a 250 kHz burst of ten cycles to J1 and watch TP15 (LST), TP16 (WAKE_N) and TP17 (BIG_N). At 20 mV the lines must stay high. At 56 mV WAKE_N must drop. At 156 mV both must drop. TP18 (REFL) must read 1.65 V all the time.

These amplitudes are at J1. The amplitudes are at SIG_IN, which J1 reaches through C33. Expect each threshold within about ±12 mV because of the comparator offset. WAKE_N and BIG_N stay low only for the length of the burst.

BENCH STEPS

1. **TP18 REFL reads 1.65 V, TP16 and TP17 read 3.3 V**
2. **Feed 250 kHz, 20 mV: both lines stay high**
3. **Feed 56 mV: WAKE_N drops, BIG_N stays high**
4. **Feed 156 mV: both drop**
5. **Remove the burst: both lines return high**

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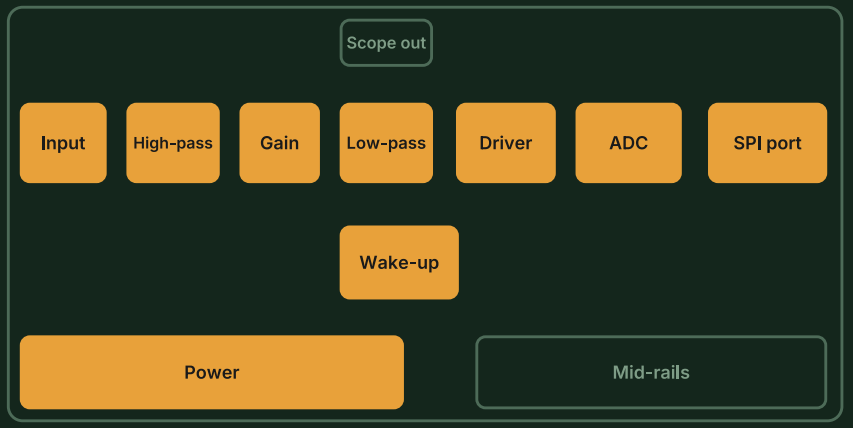
SECTION 6

The board

STEPS 36 TO 44

The stack, the floor plan, where every part sits, how the tracks and the power are routed, what the silkscreen says and which footprints need care.



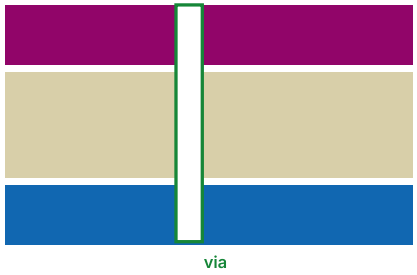
BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

36 THE BOARD
The stack and the size

The main board is 125 × 21.5 mm with two copper layers on 1.6 mm FR-4. Both layers carry a ground pour. 151 of the 246 vias are ground stitching vias that tie the two pours together.

The board has 834 track segments. 786 are 0.2 mm wide and 48 are 0.15 mm, near the fine-pitch pads. Vias are 0.6 mm with a 0.3 mm drill.

FROM THE BOARD FILE

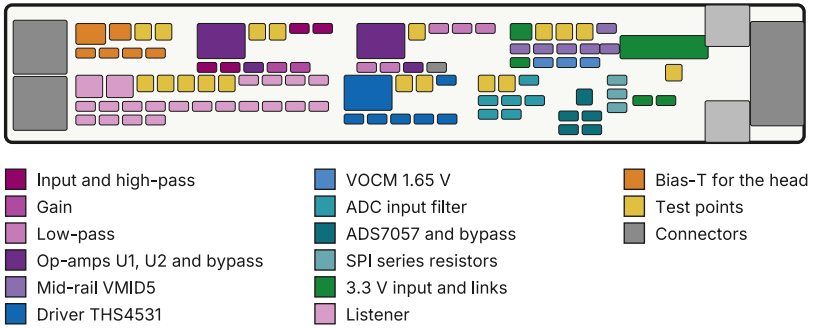


37 THE BOARD
How the floor is planned

The board is a narrow strip, 125 × 21.5 mm, so that boards fit on adjacent PMOD ports, about 22.7 mm apart. The signal runs from the SMA connectors on the left to the PMOD on the right.

The top row holds the bias-T, the high-pass, the gain, the low-pass and the mid-rail. The bottom row holds the listener, the driver, the ADC input filter and the ADC. The power header J7 sits next to the PMOD.

TOP VIEW, COLORS BY BLOCK



38 THE BOARD
Where every part is

The table lists the blocks, the references and the count. The main board has 101 footprints: 75 surface-mount parts, 4 through-hole parts and 2 mounting holes and 18 test points, plus 2 solder jumpers (JP1 and JP3) that are copper only.

Every part has a place in exactly one block, and the same reference appears in the schematic, the BOM and the placement file.

BLOCKS, REFERENCES

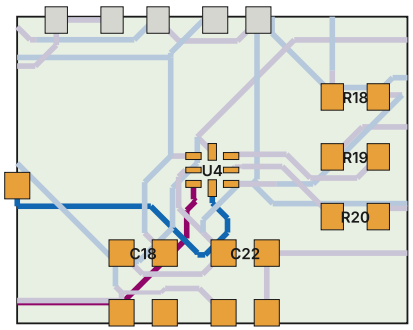
- **Input and high-pass** — C1 C2 R1 R2
- **Gain** — R3 R4
- **Low-pass** — C4 C5 C6 R5 R6
- **Op-amps U1, U2 and bypass** — C3 C10 U1 U2
- **Mid-rail VMID5** — C7 C8 C9 R7 R8 R9
- **Driver THS4531** — C11 C12 R10 R11 R12 R13 U3
- **VOCM 1.65 V** — C13 R14 R15
- **ADC input filter** — C14 C15 C16 C17 R16 R17
- **ADS7057 and bypass** — C18 C19 C20 C22 U4
- **SPI series resistors** — R18 R19 R20
- **3.3 V input and links** — C21 C23 JP3 J7 R34
- **Listener** — C26 C27 C28 C40 C41 C42 C43 C44 C45 R21 R2…
- **Bias-T for the head** — C32 C33 JP1 L2 R26 R27
- **Test points** — TP3 TP4 TP5 TP6 TP7 TP8 TP9 TP10 TP11 TP12…
- **Connectors** — J1 J3 J4 R25
- **Holes** — H1 H2

39 THE BOARD
Why the ADC corner is built this way

Everything that touches the ADC input sits around U4: the two 10 Ω resistors, the four 100 pF capacitors, the supply capacitors and the three 22 Ω resistors. The tracks AINP and AINM are 37 and 28 mm long.

A length difference of 9 mm is harmless here. A 5 mm track delays a signal by about 30 ps, and a 500 kHz wave lasts 2 µs.

REAL BOARD, ADC CORNER



40 THE BOARD
How the analog tracks are routed

The longest analog net is LPF_OUT at 39 mm, because it feeds the SCOPE OUT connector and the driver. All other analog nets are under 44 mm.

The router chose the paths, and the board is small enough that none of them matters at 500 kHz. The figure lists the lengths.

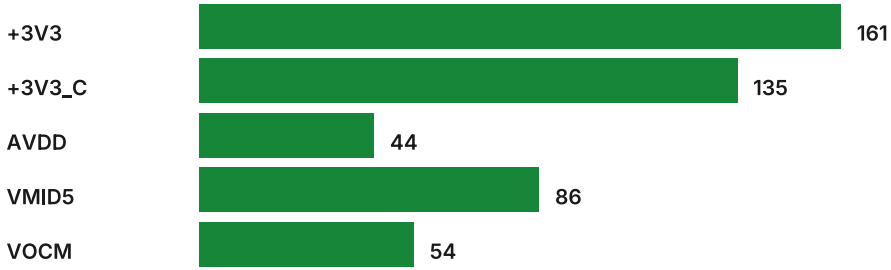
NET LENGTH, mm

41 THE BOARD
How power and ground are routed

Ground is a pour on both layers, tied by 151 vias, so every ground pad finds a short path. The power nets are 0.2 mm tracks, because the currents are small.

The +3V3_C net is 135 mm of track in total. With 27 mA (22 mA, plus 5 mA for the sensor head) this drops about 9 mV across its whole length (ESTIMATE, worst case).

POWER NET LENGTH, mm

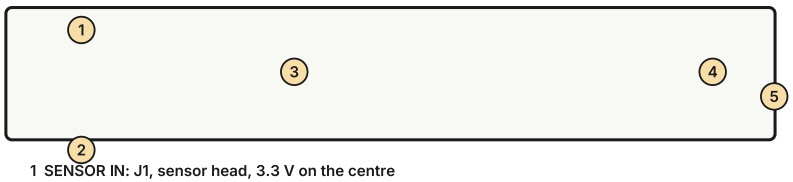


42 THE BOARD
What each label on the silkscreen means

The front carries SENSOR IN and SCOPE OUT. It also has the number of every test point (TP3 to TP20) and every IC (U1 to U4, U6, U9). It names the jumpers JP1 and JP3 and the pins of the header J7. The small parts (R16, C29) have no name on the silkscreen, because the board is dense. Their names are in the KiCad file and in the 1:1 print.

The back is one panel. It has a table of the test point names, the PMOD pins, the J7 pins and the jumpers. It also has the Auselia logo and the title AFE Test Board.

FRONT SIDE



1 SENSOR IN: J1, sensor head, 3.3 V on the centre
2 SCOPE OUT: J4, the signal at the ADC (front)
3 test points: TP and IC numbers on the top silkscreen; the table of names is on the back
4 power header J7 (1 to 5) and jumpers JP1, JP3: names on the top
5 PMOD pin list, J7 legend and the AFE Test Board logo: back

43 THE BOARD
Which footprints are tricky

The ADS7057 pads are 0.55 × 0.25 mm at 0.5 mm pitch. The gap between pads is 0.25 mm, so the solder mask web is thin. JLCPCB assembled it in the quote without a warning.

Three other parts need care. The factory model of each one had a different orientation or origin from the footprint, and the CPL now corrects it.

PARTS TO WATCH

U4 ADS7057
X2QFN-8, 1.5 mm
pin 1 mark checked, rotated 180

U6, U9 TLV7022
SOT-23-8, 0.65 mm pitch
checked in the viewer, rotation 180

U8 OPA365 (head)
SOT-23-5
rotation 0 after the viewer check

J3, J7 headers
J3 right-angle 2 × 6, J7 1 × 5
angle and centre moved in the CPL

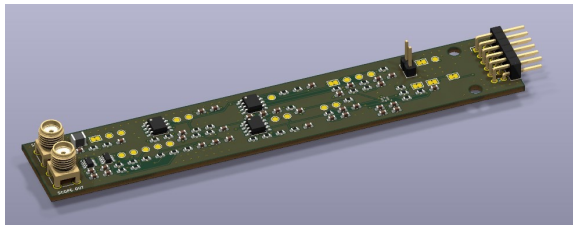
THE BOARD

44 What the board looks like

The render is from the final KiCad file. J1 and J4 stand on the left edge, J1 above J4. The PMOD header J3 points off the right edge. The 5-pin power header J7 and the solder jumpers are next to it.

The two holes are M3 mounting holes.

3D RENDER, KICAD



J1: SENSOR IN, left edge, top

J4: SCOPE OUT, left edge, bottom

J3: PMOD, right edge

J7: power and SMU header

U4: ADS7057, right third

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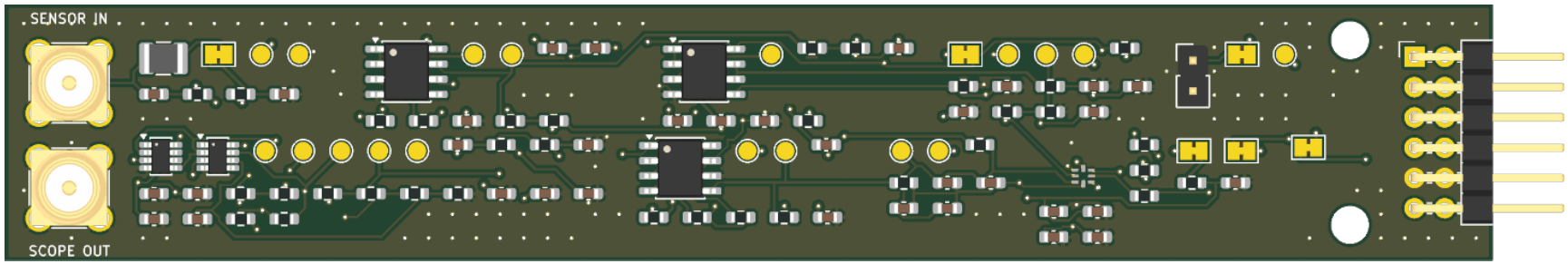
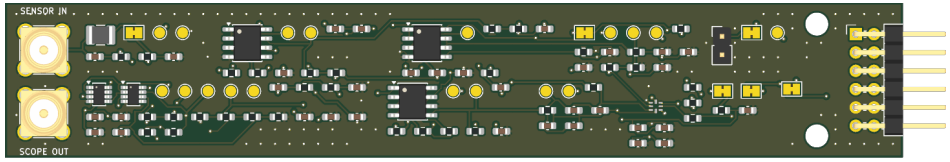


10 mm at this scale (1.8 : 1)

Board 125.0 × 21.5 mm, KiCad 3D render, view from the top



The same view at real size, 1 : 1

SECTION 7

Making the board

STEPS 45 TO 48

The files the factory needs, the parts list, what the placement viewer corrected and what the order costs.

MAKING THE BOARD

45 Which files go to the factory

JLCPCB needs three files. The Gerber zip makes the bare board. The BOM names every part. The CPL gives the position and angle of each part.

The main board zip holds 13 files. The BOM has 27 lines and 79 parts. The CPL has 79 rows. The mounting holes, the solder jumpers JP1 and JP3 and the test points are in none of them. The sensor head has its own zip, BOM and CPL.

FILES IN hw/fab

ADC_Test_gerbers.zip
Copper, mask, silk, paste, outline, drill

ADC_Test_BOM_JLC.csv
Value, references, footprint, LCSC number

ADC_Test_CPL_JLC.csv
Reference, X, Y, layer, rotation

MAKING THE BOARD

46 What is in the BOM

27 lines hold 79 parts. 5 lines hold extended parts, and JLCPCB charges a fee for each type. All other lines hold basic parts.

The ADC, the driver, the op-amps, the comparators and the 470 µH inductor of the bias-T are the extended parts.

EXTENDED PARTS, PRICE PER PART

Ref	Part	LCSC	Price
L2	470uH	C75644	$0.09
U1,U2	OPA2365	C130011	$1.38
U3	THS4531	C705347	$2.99
U4	ADS7057	C2669902	$7.98
U6,U9	TLV7022	C2862588	$0.71

Through-hole parts: J1 and J4 SMA, J7 and J3. JLCPCB solders them by hand, for a fee per joint.

MAKING THE BOARD

47 What the placement viewer corrected

The JLCPCB viewer showed parts turned wrong on the first revision. 7 rotations are corrected in the CPL. The new parts of this revision (U6 and U9 TLV7022, U8, L1, L2, J6) are not checked in the viewer yet. Each fix sits in `jlc_files.py`, not in the board.

The KiCad footprints and the JLC library disagree on the zero angle. J3 and J7 have their origin on pin 1. So the CPL moves both to the centre of the part. J7 is turned to 0, U6 and U9 to 180, and U8 on the head to 0.

KICAD ROTATION AGAINST CPL ROTATION

Ref	KiCad	CPL	Why
J3	0°	90°	viewer showed it turned; set absolute angle
J7	90°	0°	viewer showed it turned; set absolute angle
U1	0°	270°	viewer showed it turned; set absolute angle
U2	0°	270°	viewer showed it turned; set absolute angle
U3	0°	270°	viewer showed it turned; set absolute angle
U6	0°	180°	viewer showed it turned; set absolute angle
U9	0°	180°	viewer showed it turned; set absolute angle

Rule: look at every part in the viewer once. The viewer shows the real answer.

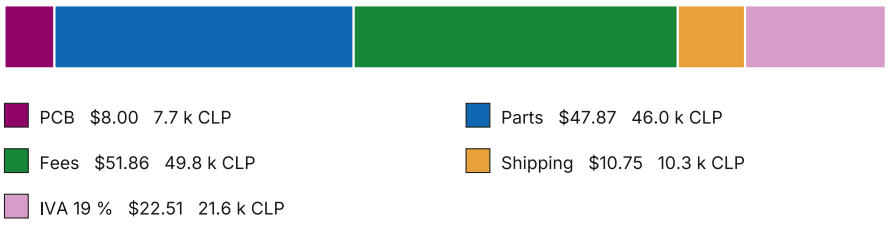
MAKING THE BOARD

48 What the order costs

Two ADC boards and five sensor heads are two JLCPCB orders. They cost $107.72 at the factory. With one parcel, shipping and IVA, the total is $140.98, or 135 thousand CLP at 960 CLP per USD.

The ADC order is $66.96 and the head order is $40.77. Five ADC boards would cost $118.11. These are estimates from the model of the first quote, until you upload the files.

ESTIMATE, 2 ADC BOARDS + 5 HEADS, USD



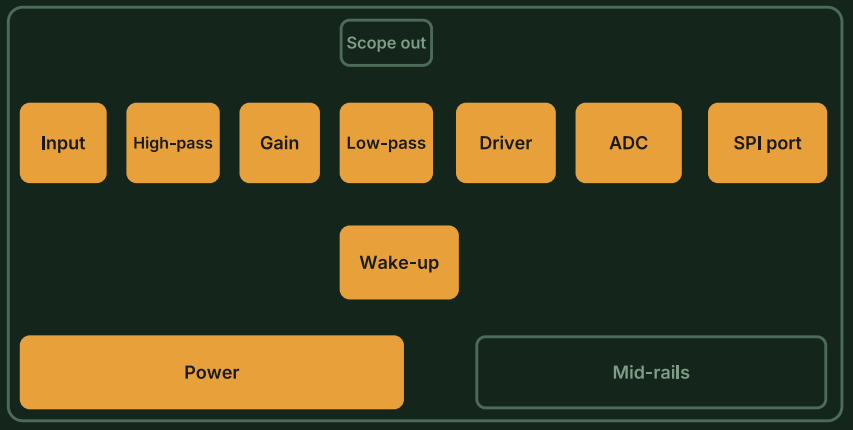
Total $140.98. Two parcels $153.78. 5 + 5 boards $201.86.

SECTION 8

Bring-up and checks

STEPS 49 TO 54

What to measure before the PMOD goes in, the first signal test, the response to expect, how to test the wake-up, what to do when something fails and what is still open.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

BRING-UP AND CHECKS

49 What to check before the PMOD goes in

Remove the J7 jumper. Feed 3.3 V to J7 pins 2 and 5 from an SMU or a bench supply with a 100 mA limit. Use pins 3 and 4 for Sense. Leave the PMOD unplugged. Measure each rail at the test points against GND. Switch off if +3V3 is off by more than 0.1 V or the current is above 60 mA.

With the PMOD plugged in and the jumper on J7 pins 1 and 2, the same readings must appear. Never use the jumper and the SMU at the same time.

EXPECTED VOLTAGES, 3.3 V IN

Node	Test point	Expect
+3V3	TP11	3.30 V
AVDD	TP12	3.30 V
VMID5	TP13	1.65 V (half of +3V3_C)
VOCM	TP14	1.65 V (half of AVDD)
REFL	TP18	1.65 V
LST	TP15	about 1.65 V
WAKE_N, BIG_N	TP16, TP17	3.3 V (pulled up)

BRING-UP AND CHECKS

50 The first signal test

Plug the PMOD in. Put 100 mV peak at 250 kHz on J1 from the Analog Discovery 2 and send a burst of ten cycles. Look at SCOPE OUT and read the SPI words.

The scope shows 410 mV peak above 1.6 V. The ADC sees 410 mV peak between its inputs, which is 1018 codes.

EXPECTED, 100 mV PEAK AT J1

Where	Expect
J1	100 mV peak, 250 kHz
SCOPE OUT (J4)	410 mV peak on 1.6 V
ADC input, differential	410 mV peak
Code, peak	about 1018 (0x03FA)
Code, no signal	near 0 (0x0000), ± 1

BRING-UP AND CHECKS

51 What response to measure

Sweep J1 at 100 mV and read the code amplitude. Divide by the amplitude at 250 kHz. The numbers below come from the simulation.

Component spread moves the 10 kHz point. In 11 of 150 random runs the loss at 10 kHz was below 40 dB.

SIMULATION, NOMINAL PARTS

Frequency	Gain SMA to ADC	Against 250 kHz
10 kHz	-28.2 dB	-40.5 dB
50 kHz	-0.5 dB	-12.7 dB
100 kHz	+9.2 dB	-3.1 dB
250 kHz	+12.3 dB	+0.0 dB
500 kHz	+9.4 dB	-2.8 dB
1000 kHz	+0.1 dB	-12.1 dB
2000 kHz	-11.7 dB	-24.0 dB

BRING-UP AND CHECKS

52 How to test the listener thresholds

Sweep a 250 kHz burst at J1 from 10 mV upward. WAKE_N must drop between about 23 and 46 mV. BIG_N must drop between about 99 and 123 mV. The table gives the three points to check.

The quiet perfboard noise is 4.6 mV rms in the band, so WAKE_N at 35 mV is about 8 times the noise. The comparator offset (up to ±8 mV, datasheet) moves each threshold by that much.

COMPUTED FROM THE RESISTORS

Burst at J1	WAKE_N	BIG_N
20 mV peak	high	high
45 mV peak	low	high
156 mV peak	low	low

ESTIMATE: the listener filters pass 0.68 at 250 kHz, from the netlist simulation.

BRING-UP AND CHECKS

53 What if it does not work

Find the first stage that is wrong, then look only at that stage. Work from the input toward the ADC. Measure DC first, then the signal.

Most first-board failures are a short, a part turned the wrong way or a missing 3.3 V.

SYMPTOM, CAUSE, CHECK

Symptom	Likely cause	Check
3.3 V low	Weak PMOD supply or short	TP11 under load
SCOPE OUT flat	U1 or U2 not powered	3.3 V at pin 8, JP3 closed
SDO all zeros	First frame only	Read twice
SDO stuck	No 3.3 V on DVDD	+3V3 at U4 pin 4
WAKE_N always low	LST noisy or REFL off	TP18 must be 1.65 V
Code offset	VOCM or VMID off	Step 49 table, JP3 closed

BRING-UP AND CHECKS

54 What is verified and what is open

Checked by tools: the electrical rules, the design rules, the netlist against the board, and the rotations in the JLC viewer. Checked by simulation: gain, band, noise and settling.

Nothing has run on a real board yet. Red rows mark the three open items. The THS4531 stability with its 400 pF load stays on the list of the ADC input filter, step 18.

STATUS

Item	Status	How
ERC, DRC, parity	Done	KiCad, 0 errors
Rotations	Done	JLC viewer
Chain response, noise	Simulated	ngspice, vendor models
Head and bias-T at 3.3 V	Simulated	ngspice, OPA365
Listener on real captures	Simulated	Python, 25 captures
As-built netlist, rails to listener	Simulated	ngspice, netlist_verify
ZedBoard 3.3 V rail noise	OPEN	probe TP11 and TP12
PMOD fit, adjacent ports	OPEN	print 1:1 and test
First board, bench	OPEN	steps 49 to 52

NOTES

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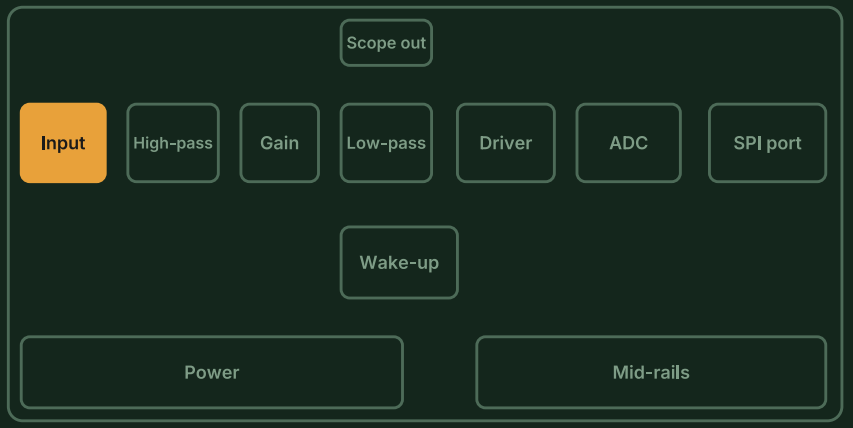
SECTION 9

The sensor head

STEPS 55 TO 60

The perfboard charge amplifier on its own small board, how it is powered through the coax by a bias-T, how it is laid out and how to test it.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

55 THE SENSOR HEAD
What the sensor head does

The head is the perfboard on a PCB. The piezo wires solder to it. It turns the charge of the piezo into a voltage and sends that voltage down one coax cable.

The same coax brings 3.3 V from the ADC board to the head. So the head needs one cable and no battery. The OPA365 is the lab default. For the battery product, fit an OPA314 (150 µA) and Cf 220 pF on the same pads.

CHAIN

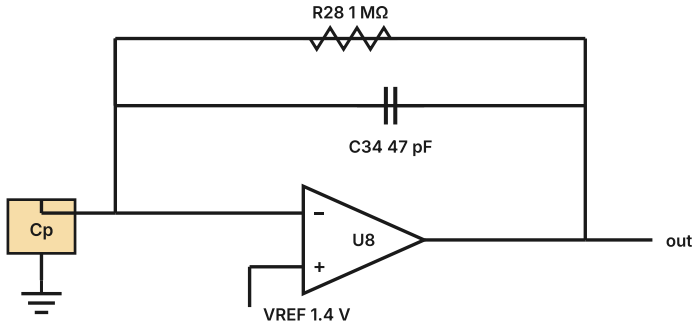
The head loses 0.17 dB at 250 kHz to the bias-T. To use a generator or a perfboard instead, cut the solder jumper JP1.

56 THE SENSOR HEAD
How the charge becomes a voltage

The amplifier keeps its input at a fixed voltage. So the charge Q of the piezo flows into the feedback capacitor Cf, and the output moves by Q / Cf. With Cf = 47 pF, a charge of 4 to 7 pC gives 85 to 149 mV.

Rf = 1 MΩ drains the capacitor with a time constant of 47 µs, a corner at 3.4 kHz. The noise gain is 35, so the 50 MHz amplifier keeps 1.4 MHz of bandwidth.

FROM THE NETLIST, ESTIMATE FOR Q

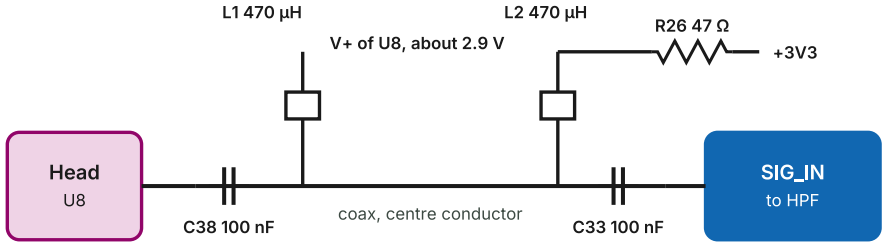


57 THE SENSOR HEAD
How the coax carries power and signal

A bias-T joins a DC path and an AC path on one wire. The DC path is an inductor: 470 µH looks like 738 Ω at 250 kHz, so it hardly loads the signal. The AC path is a capacitor: 100 nF passes 250 kHz and blocks the DC.

The simulation with the OPA365 model gives a loss of 0.17 dB at 250 kHz and 19 dB at 10 kHz. The low-frequency noise of the perfboard is cut before it reaches the filters.

SIMULATION, RG174 1 M



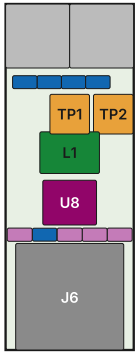
L1 and L2 are the same part. Cut the solder jumper JP1 (between C32 and L2) to turn the 3.3 V off.

58 THE SENSOR HEAD
Where the parts are on the head

The head is 10 × 27 mm, with 16 parts, all on one face. The piezo sits on the other face, flat against the pads. The SMA stands on the parts face and points out of the capsule.

The capsule drawing v3 has a PCB of 10 × 20 mm. With the SMA and the inductor the head needs 2 mm more, so the PCB sticks out 9 mm instead of 7.

REAL BOARD, TOP VIEW, 2.2 : 1



U8	OPA365, charge amplifier
L1	470 µH, supply from the coax
TP1	piezo +, summing node
TP2	piezo –, ground
J6	SMA, vertical
R28 C34	Rf 1 MΩ, Cf 47 pF
R29 R30 C35	VREF 1.4 V
C36 C37	supply bypass
R31 C38	49.9 Ω and 100 nF out

THE SENSOR HEAD

59 What goes to the factory for the head

The head has its own Gerber zip, BOM and CPL. It is 8 BOM lines and 12 parts. The OPA365 and the inductor come from the extended library.

Five heads cost about $40.77 at the factory, $8.15 each (ESTIMATE). Fixed fees are most of it: setup, stencil and hand soldering of the SMA.

HEAD BOM, PRICE PER PART

Ref	Part	LCSC	Price
C34	47pF C0G	C1567	$0.008
C35,C37	10uF	C15525	$0.026
C36,C38	100nF	C307331	$0.010
J6	SMA jack	C496549	$0.402
L1	470uH	C75644	$0.090
R28,R29,R30	1M	C26083	$0.002
R31	49.9R	C25120	$0.002
U8	OPA365	C96626	$2.674

THE SENSOR HEAD

60 How to test the head

Power the ADC board from the PMOD (jumper on J7 pins 1 and 2) or from an SMU on J7. Connect the head to SENSOR IN (J1) with the coax. Measure the head first, then inject a known charge. A 10 pF capacitor from the function generator to the piezo + pad gives Q = 10 pF × ΔV.

A step of 100 mV injects 1 pC. The output moves by 21 mV, because Q / Cf = 1 pC / 47 pF.

EXPECTED VALUES

Where	Expect
C37 (V+ of U8)	about 2.9 V
VREF, C35	about 1.4 V after 0.2 s
U8 output, R31 pad 1	equal to VREF
J1 centre pin	about 3.0 V
Extra current on the 3.3 V input	about 4 mA
100 mV step via 10 pF	21 mV step at SIG_IN

NOTES

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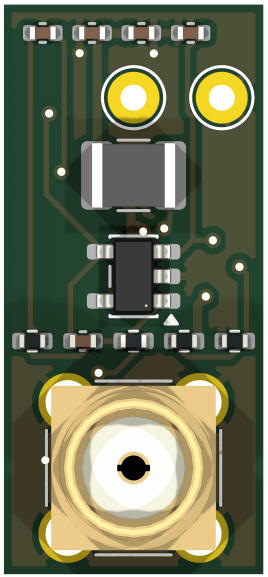
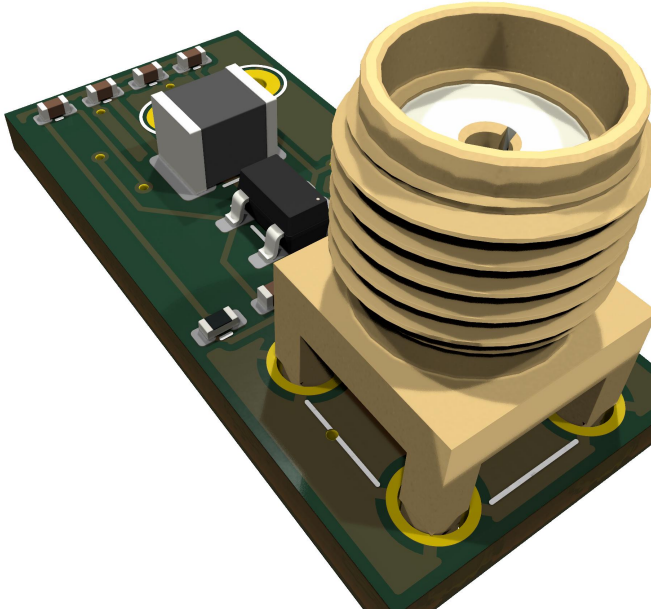
FRONT, THE PARTS FACE

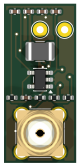


BACK, THE PIEZO FACE

Both at 3.6 : 1, board 10 × 27 mm. The + pad is the centre of the piezo.



Seen at an angle, from the parts face.



Real size, 1 : 1

SECTION 10

Why is it like this?

STEPS 61 TO 69

Nine questions that come up when you study the board: the differential head, the supply filter, the buffer, the gain, Q, noise density, power-up order, open-collector outputs and the test points.

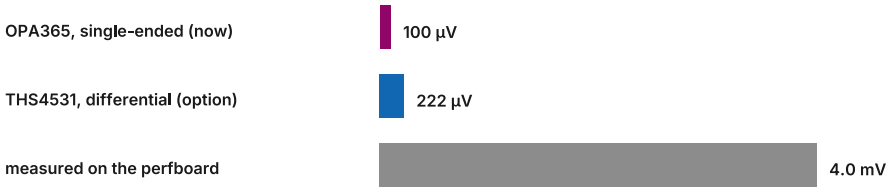
QUESTIONS AND ANSWERS

61 Why is the head not differential?

A differential amplifier subtracts its two inputs. Pickup that hits both wires equally cancels. That works only if both faces of the piezo float. Today the minus face connects to ground, so the head is single-ended.

The THS4531 would add noise: 222 µV against 100 µV at the output in the AE band (ESTIMATE). It also draws 160 nA of input bias current, which a 1 MΩ resistor turns into 0.16 V. The measured noise is 4 mV, so only a test shows if the wires or the plant cause it.

NOISE AT THE HEAD OUTPUT, 100 TO 500 kHz



Test: tie both piezo faces to a differential input and compare the noise with the plant touching the stem.

QUESTIONS AND ANSWERS

62 Is the 3.3 V supply clean enough?

The ZedBoard makes 3.3 V with a switching regulator, so its ripple can sit in the band the sensor listens to. AVDD is the ADC reference, so it gets a filter: a 10 Ω resistor (R34) in series and 10 µF (C21) to ground.

With the estimates in the table, the ripple on AVDD falls by 39 dB at 250 kHz. The op-amps reject the rest of the supply by their own PSRR. The real ZedBoard noise is not known, so probe TP11 and TP12 with 20 MHz bandwidth before you trust it.

ESTIMATE, ASSUMED PARTS

assumed

R34 10 Ω

C21 10 µF at 3.3 V: 6 µF

ESR 10 mΩ

100 kHz	250 kHz
-32 dB	**-39 dB**
500 kHz	**1000 kHz**
-45 dB	**-51 dB**

Measure: 20 MHz bandwidth, AC coupling, 10 mV per division at TP11 and TP12.

QUESTIONS AND ANSWERS

63 Why copy the mid-rail with U2B?

A divider cannot supply current. The two 10 kΩ resistors look like 5 kΩ. The gain stage returns up to 0.12 mA of signal current into VMID5. Across 5 kΩ that would move the rail by 0.6 V.

The capacitors C8 and C9 hold the rail in the AE band: 63 mΩ at 250 kHz. The buffer matters below a few kHz, where the capacitors are weak. It costs nothing, because it uses the spare half of U2.

IMPEDANCE OF VMID5, Ω

frequency	capacitors alone	with the buffer
10 Hz	1.58e+03 Ω	**10 Ω**
100 Hz	158 Ω	**10 Ω**
1 kHz	15.8 Ω	**10 Ω**
10 kHz	1.58 Ω	**1.58 Ω**
100 kHz	0.158 Ω	**0.158 Ω**
250 kHz	0.063 Ω	**0.063 Ω**

Buffer path: U2B, R9 10 Ω and C8. Capacitors: C8 10 µF and C9 100 nF. Divider: 5 kΩ.

QUESTIONS AND ANSWERS

64 Why a gain of 4.9?

The head gives 85 to 149 mV for a charge of 4 to 7 pC. With no gain, a 120 mV event reads about 262 codes of 16384, 1.5 % of full scale. The sensor noise is 4 mV rms, so the ADC noise of 0.6 LSB (0.24 mV) hardly matters in either case.

The gain is not needed for the noise. It makes the event fill about 5 times more codes and it follows the AFE budget. The price is headroom: the stages clip at a smaller input. R3 sets the gain, so a change is one resistor.

SAME 120 mV EVENT, THREE GAINS

Gain	R3	codes at the ADC	clips at the head
x1	wire	262	1694 mV
x2	1 kΩ	524	847 mV
x4.9	3.9 kΩ	1284	346 mV

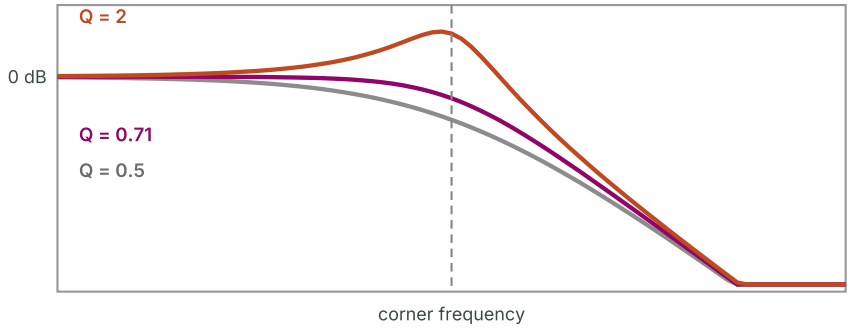
QUESTIONS AND ANSWERS

65 What is Q?

Q tells how much a second-order filter peaks near its corner frequency. A low Q gives a soft corner. A high Q gives a peak and ringing. Q = 0.71 has no peak and is the flattest curve, the Butterworth response.

Both filters here use Q = 0.71. In the time domain, Q = 0.71 overshoots a step by 4 %. Q = 2 overshoots by 44 %.

SECOND-ORDER LOW-PASS



QUESTIONS AND ANSWERS

66 What is nV/√Hz?

Noise voltage spreads over frequency. A density of 4.5 nV/√Hz means 4.5 nV rms in each 1 Hz slice. The slices are independent, so their powers add, not their voltages. The rms of a band is the density times the square root of the bandwidth.

The OPA365 has 4.5 nV/√Hz. In the AE band of 400 kHz that is 2.8 µV rms. The charge amplifier has a noise gain of 35, so 100 µV reach its output.

OPA365, 4.5 nV/√Hz

1 Hz	4.5 nV × √1	**4.5 nV**
1 kHz	4.5 nV × √1000	**142 nV**
100 kHz	4.5 nV × √100000	**1.42 µV**
400 kHz, AE band	4.5 nV × √400000	**2.85 µV**

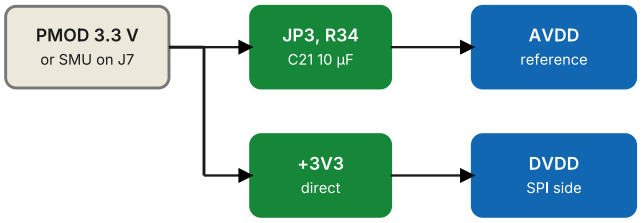
QUESTIONS AND ANSWERS

67 Does the power-up order matter?

The ADS7057 datasheet gives no sequencing rule. It gives limits: AVDD and DVDD up to 3.9 V, inputs up to AVDD + 0.3 V and logic inputs up to DVDD + 0.3 V. The 22 Ω resistors and the 10 Ω resistors keep a stray current under the 10 mA pin limit.

On this board AVDD and DVDD come from the same 3.3 V rail, so they rise together. One thing matters in practice: AVDD must be stable before the first SPI frame, because that frame calibrates the offset. The VMID5 rail needs about 25 ms (step 4), so wait that long after power-up.

WHO POWERS WHAT



Ground is common. Keep the SPI at 10 to 20 MHz while you check the noise.

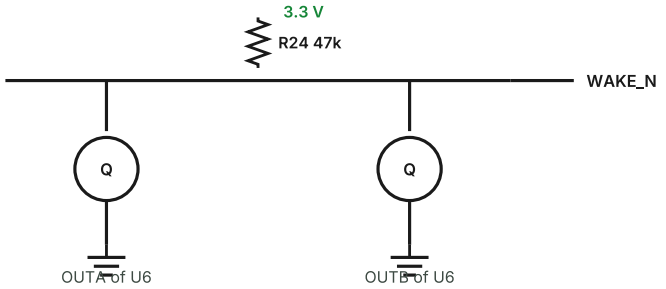
QUESTIONS AND ANSWERS

68 What is an open-drain output?

The output stage of the TLV7022 is one transistor. It can pull the output to ground. It cannot push it high. When it is off, the pin floats.

A resistor to 3.3 V (R24 or R42, 47 kΩ) supplies the high level. Two outputs on one wire make an OR for low: either one pulls the line low. The pull-up also sets the voltage level, so the FPGA never sees more than 3.3 V.

TWO OUTPUTS ON ONE WIRE



both off: high. either on: low.

QUESTIONS AND ANSWERS

69 Do the test points disturb the circuit?

Slightly. Each test point is a pad and a stub of track, about 0.5 pF per centimetre. The stubs here are 4 to 12 mm long, so 1 to 2 pF each. A 10:1 scope probe adds 10 pF, more than any stub, so the probe changes the circuit more than the test point does.

The stubs sit on low-impedance nodes: op-amp outputs, or the ADC input behind 400 pF. The risk is pickup on a long stub. If the noise is too high, cut the stub of the quiet stage with a knife.

STUB LENGTH AND CAPACITANCE, ESTIMATE

SIG	9 mm	**0.8 pF**	**DRV+**	12 mm	**0.9 pF**
HPF	9 mm	**0.7 pF**	**DRV-**	6 mm	**0.6 pF**
GAIN	6 mm	**0.6 pF**	**AIN+**	5 mm	**0.6 pF**
LPF	7 mm	**0.6 pF**	**AIN-**	4 mm	**0.5 pF**

One-off check: probe a stage with and without the 10 pF of the probe and compare.

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WHERE THINGS LIVE

Files, models and datasheets

`hw/`	KiCad project, schematic, both boards
`hw/fab/`	Gerbers, BOM and CPL for JLCPCB, both boards
`hw/tools/`	Generators, DRC, cost model, 1:1 print
`sim/`	Chain simulation, vendor models list, LTspice netlist
`doc/tools/`	This manual, built from the files above
`doc/data/`	Board and simulation snapshots used in the figures

DATASHEETS

ADS7057, THS4531, OPA365 (OPA2365), TLV7022

```
Rebuild: M4_PDF_PYTHON=<python with pymupdf> python3 doc/tools/make_adc_manual.py
```